Python Programming - 2301CS404

Lab - 3

01) WAP to check whether the given number is positive or negative.

In [3]:
```
num = int(input("Enter The Number: "))
if num > 0:
    print("Number Is Positive")
elif num < 0:
    print("Number Is Negative")
else:
    print("Number Is Zero")
```

```
Number Is Zero
```

02) WAP to check whether the given number is odd or even.

In [5]:
```
num = int(input("Enter The Number: "))
if (num % 2 == 0):
    print("Number Is Even")
else:
    print("Number Is Odd")
```

```
Number Is Odd
```

03) WAP to find out largest number from given two numbers using simple if and ternary operator.

In [45]:
```
# num1 , num2 = input("Enter The Numbers").split()
num1 = int(num1)
num2 = int(num2)

print(f"Largest Is {num1}") if num1 > num2 else print(f"Largest Is : {num2}")
```

```
20
```

04) WAP to find out largest number from given three numbers.

In [47]:
```
num1 = int(input("Enter The First Number"))
num2 = int(input("Enter The Second Number"))
num3 = int(input("Enter The Third Number"))

print(num1) if num1 > num2 & num1 > num3 else print(num2) if num2 > num3 else pr
```

```
50
```

05) WAP to check whether the given year is leap year or not.

[If a year can be divisible by 4 but not divisible by 100 then it is leap year but if it is divisible by 400 then it is leap year]

In [60]:
```
year = int(input("Enter the Year: "));

print("Year Is Leap") if ((year%4==0 and year%100!=0) or year%400 == 0) else pri
```

```
Year Is Not Leap
```

06) WAP in python to display the name of the day according to the number given by the user.

In [69]:
```
day = input("Enter The Day: ")
day= int(day)
days= ["" ,"Monday","Tuesday","Wednesday","Thursday","Friday","Saturday","Sunday
print(days[day])
```

```
Sunday
```

07) WAP to implement simple calculator which performs (add,sub,mul,div) of two no. based on user input.

In [74]:
```
print("1.Add \n2.Sub \n3.Mul \n4.Div")
choice = int(input("Enter The Choice:"))
a = int(input("Enter The First Number: "))
b = int(input("Enter The Second Number: "))
match(choice):
    case 1:
        print(a + b)
    case 2:
        print(a - b)
    case 3:
        print(a * b)
    case 4:
        print(a // b)
```

```
1.Add
2.Sub
3.Mul
4.Div
84
```

08) WAP to read marks of five subjects. Calculate percentage and print class accordingly.

Fail below 35

Pass Class between 35 to 45

Second Class

between 45 to 60

First Class between 60 to 70

Distinction if more than 70

In [101…

```
sub1 = int(input("Enter marks for subject 1: "))
sub2 = int(input("Enter marks for subject 2: "))
sub3 = int(input("Enter marks for subject 3: "))
sub4 = int(input("Enter marks for subject 4: "))
sub5 = int(input("Enter marks for subject 5: "))
total_marks = sub1 + sub2 + sub3 + sub4 + sub5
percentage = (total_marks / 500) * 100

print(f"Total Marks: {total_marks}")
print(f"Percentage: {percentage:.2f}%")

if percentage > 70:
    print("Class: Distinction")
elif percentage >= 60:
    print("Class: First Class")
elif percentage >= 45:
    print("Class: Second Class")
elif percentage >= 35:
    print("Class: Pass Class")
else:
    print("Class: Fail")
```

```
Total Marks: 460
Percentage: 92.00%
Class: Distinction
```

09) WAP to find the second largest number among three user input numbers.

In [100…

```
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))
num3 = float(input("Enter the third number: "))

if (num1 >= num2 and num1 <= num3) or (num1 >= num3 and num1 <= num2):
    second_largest = num1
elif (num2 >= num1 and num2 <= num3) or (num2 >= num3 and num2 <= num1):
    second_largest = num2
else:
    second_largest = num3

print(f"The second largest number is: {second_largest}")
```

```
The second largest number is: 23.0
```

10) WAP to calculate electricity bill based on following criteria. Which takes the unit from the user.

a. First 1 to 50 units – Rs. 2.60/unit

b. Next 50 to 100 units – Rs. 3.25/unit

c. Next 100 to 200 units – Rs. 5.26/unit

d. above 200 units – Rs. 8.45/unit

In [90]:
```
unit = int(input("Enter THe Unit"))

if unit <= 50:
    print(f"Rs. 2.60/unit \nTotal Price {unit*2.60}")
elif unit <= 100:
    print(f"Rs. 3.25/unit \nTotal Price {unit*3.25}")
elif unit <= 150:
    print(f"Rs. 5.26/unit \nTotal Price {unit*5.26}")
else:
    print(f"Rs. 8.45/unit \nTotal Price {unit*8.45}")
```

```
Rs. 2.60/unit
Total Price 2.6
```

In [ ]: